

Machine Learning Applications: Housing and Autonomous Transportation

1. Traditional Machine Learning: California House Price Prediction

During a previous internship, I worked on a California house price prediction project. The goal was to estimate a property's market value using structured information such as location, square footage, number of bedrooms and bathrooms, property age, and neighborhood characteristics. Similar valuation systems are used by real estate platforms to help buyers and sellers understand approximate home values. For example, Zillow reports that its Zestimate system analyzes hundreds of data points for individual properties.

Traditional machine learning is suitable for this problem because most housing information can be organized into clearly defined rows and columns. Regression models, decision trees, random forests, or gradient boosting models can learn the relationships between these property features and historical selling prices. These models are also relatively easy to train, evaluate, and explain to a client. For example, the results may show that location, living area, and recent neighborhood sales have the greatest influence on the predicted price.

Deep learning would not be my first choice for this version of the problem. Neural networks usually require more training data, computing resources, and tuning. They may also be harder to explain to real estate professionals or customers. Deep learning could become useful if the project included property photographs, satellite images, or large amounts of unstructured text, but it would add unnecessary complexity when the main dataset is structured.

2. Deep Learning: Autonomous Taxis

The deep learning application that interests me most is autonomous driving, especially autonomous taxi services. Waymo, for example, operates a public, fully autonomous ride-hailing service.

Deep learning is suitable because an autonomous vehicle must interpret large amounts of complex, unstructured sensor data. Cameras, LiDAR, radar, and other sensors continuously collect information about pedestrians, vehicles, traffic lights, road markings, and unexpected obstacles. Deep neural networks can recognize objects, estimate their movement, understand the surrounding environment, and support real-time driving decisions.

Traditional machine learning alone would not be suitable for the complete autonomous-driving problem because engineers cannot manually create enough rules or simple features for every possible road situation. A traditional model might work for an individual task, such as predicting travel time, but it would struggle to understand raw images and complicated traffic scenes. Deep learning can automatically learn visual and spatial patterns from large datasets.

However, deep learning also creates important concerns. Its decisions can be difficult to explain, and unusual situations may not be represented well in the training data. For autonomous taxis, a prediction error can affect human safety. Therefore, the system requires extensive testing, simulation, monitoring, redundant sensors, and human-designed safety controls.

3. Conclusion

These examples show that the most advanced method is not always the most appropriate one. Traditional machine learning is practical and explainable for structured house-price data, while deep learning is more appropriate for the complex sensory information involved in autonomous driving. The correct choice depends on the data, business goal, available resources, need for explainability, and consequences of an incorrect prediction.